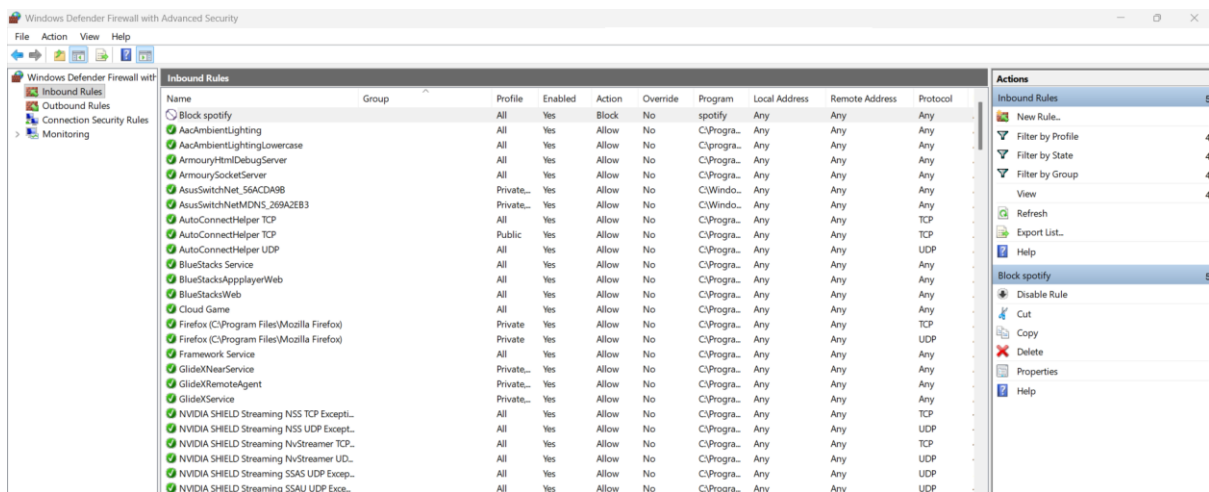


Day 4: Firewall Basics Theory (1 hour):Firewall types and use cases (hardware and software firewalls).**Practical (1 hour):**Configure basic rules in a software firewall (e.g., Windows Defender).**Task: Document how to block and allow specific traffic using a firewall.**

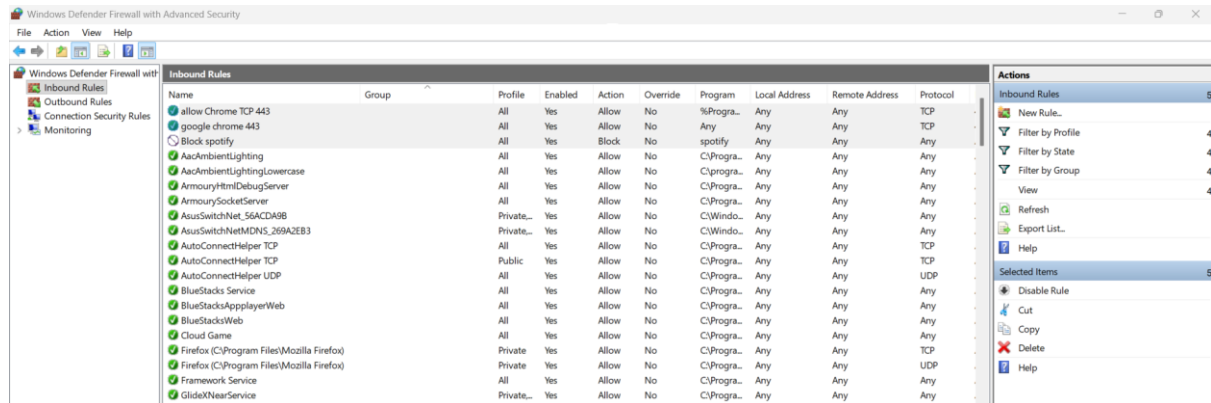
Task 1.

Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.



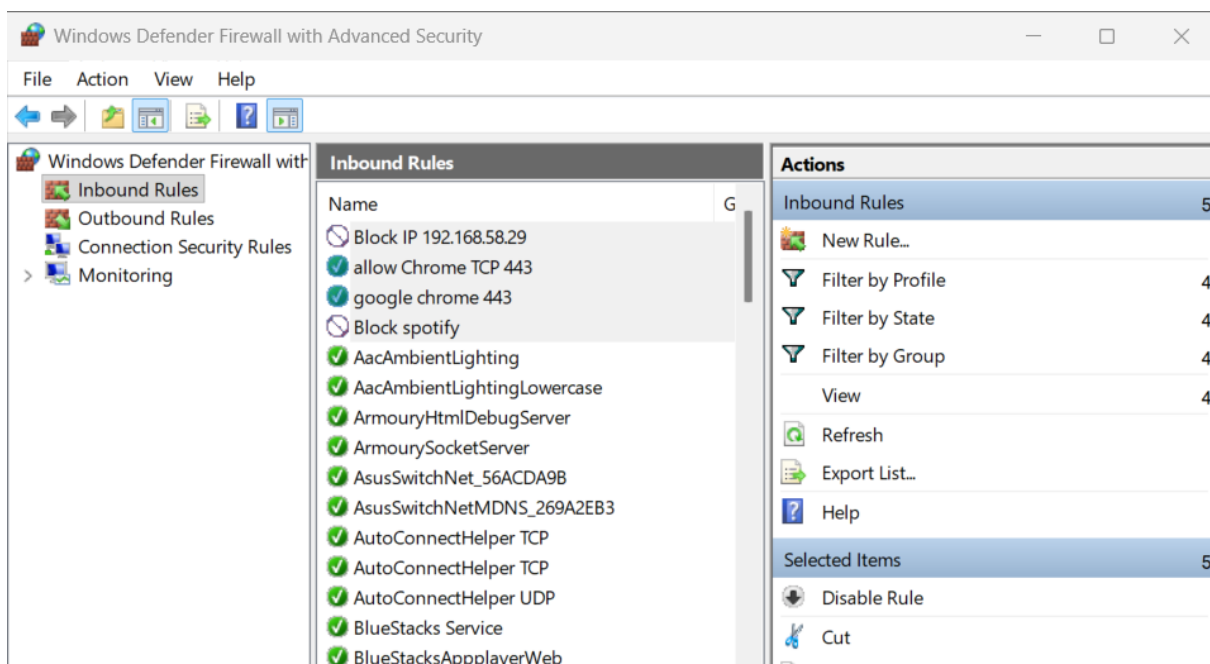
Task 2.

Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).



Task 3.

Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.
Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.



Task 4.

Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Feature	Hardware Firewall	Software Firewall
Location	Separate physical device, usually placed between the network and Internet	Installed/running on a computer or server
Protection	Can protect many devices on an entire network	Mainly protects the individual computer
Performance	Designed to handle large amounts of network traffic	Uses the computer's CPU/RAM resources
Example	Enterprise firewall/router used in an office	Windows Defender Firewall
Typical use case	Large office, company, school, or data center	Personal computer or individual server

key differences

1. Coverage

- Hardware firewall → protects an entire network.
- Software firewall → primarily protects one device.

2. Deployment

- Hardware firewall → separate physical/network appliance.
- Software firewall → installed and configured on the operating system.

3. Performance and scalability

- Hardware firewalls are designed for high-volume traffic and many users.
- Software firewalls are convenient for individual systems but depend on the host computer's resources.

Use cases

Hardware firewall:

A company with 100 employees can place a hardware firewall between its Internet connection and internal network to control traffic for all employees.

Software firewall:

A student can use Windows Defender Firewall on a Windows laptop to control which applications are allowed to communicate over the network.